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HESA-A: new treatment for breast cancer
and choroidal metastasis

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Summary

Background:

Breast cancer is one of the main causes of mortality among women worldwide. This type of cancer metastasizes to different body tissues, giving rise to many problems. The effect of HESA-A, a drug of herbal-marine origin, on vision, quality of life, and survival of end-stage breast cancer patients was investigated in this study.

Material/Methods:

In a double-blind study, 24 breast cancer patients with choroidal metastasis, aged between 41 and 49 years, were divided into case and control groups, treated with 50 mg/kg/day of HESA-A and placebo, respectively. The patients were evaluated in respect to the intensity of experienced pain, by assessing their rate of narcotic analgesic use. The patients' vision scale was also evaluated.

Results:

Notable improvement was seen in the vision of patients treated with HESA-A. Patients receiving HESA-A used narcotics at lower doses, owing to reduced experience of pain. No changes were observed in the vision of control group patients, or their pain experience.

Conclusions:

The effects of natural compounds with antioxidant and anticancer properties have been emphasized by different studies. HESA-A is a compound of natural origin, consisting of rare elements and organic materials, which in several animal and cellular studies has shown powerful anticancer effects and less toxicity on normal cells. The results of this study showed considerable improvement in the vision of breast cancer patients treated with HESA-A.

key words:

breast cancer • choroidal metastasis • HESA-A

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BACKGROUND

Cancer is a complicated disease, with each individual tumor exhibiting to a variable degree a set of overlapping phenotypes that include abnormal proliferation and cell survival, genetic instability, tissue invasion, and metastasis [1,2]. Malignant tumors from other parts of the body can spread to areas in and around the eye. These tumors may never be discovered unless they affect the vision, or push the eye forward.

The vascular choroidal tissue of the eye is a common site of involvement in metastatic cancer (choroidal metastasis). Metastatic cancers appearing in and around the eye usually originate from cancer of the breast (in women) or the lungs (in men). The source of choroidal metastasis is not clear in some patients. Early treatment offers the best hope for preserving vision, providing the center of the retina has not been destroyed by the metastatic tumor.

While any part of the eye can be involved by metastatic disease, the most commonly involved tissue is the highly vascular choroid [3,4]. Breast cancer can metastasize to bone tissue [5], macula [6], optic disk [7], optic nerve [8], and choroidal tissue [9].

Besides pain, visual impairment may affect the lives of cancer patients with eye metastases. To date, no drug has been shown to have an obvious and categorical effect in the treatment of various types of cancer [10].

The use of antioxidants had been shown to render some protection against cancers. The toxicity of chemical anticancer drugs is a major limiting factor in their application; hence the use of natural drugs, especially compounds containing selenium (Se), zinc (Zn), and nickel (Ni), has for some time been the focus of different studies [11, 12].

HESA-A is a drug of herbal-marine origin, which contains elements such as Br, Sr, Va, Ti, Mn, Ni, Ag, Cu, Zn, Cs, and Cl. Some studies have demonstrated the antitumor properties of some of these elements [13]. The anticancer effects of HESA-A has been the subject of both *in vivo* and *in vitro* studies [14,15].

The effects of HESA-A on the vision of end-stage breast cancer patients with optic metastasis was investigated in this study.

MATERIAL AND METHODS

This was a double-blind clinical trial of 24 end-stage breast cancer patients with optic metastasis, ranging in age from 41 to 49 years (mean age 45.3 years). The patients were selected at the Cancer Research Center of the Tehran University of Medical Sciences, and were divided into case and control groups equal in number. They had severe skeletal pain and reduction of eyesight. Bone metastasis in sternum, pelvis, extremities and retina choroid were recognized upon clinical examination and fluorescing angiography. Due to the advanced stage of cancer in these patients, various modes of treatment, including radiotherapy and chemotherapy, were deemed to be of little benefit. The inclusion criteria were

- metastatic breast cancer;
- more than 50% reduction of eyesight, as determined by a specialist and patient history;

- lack of any therapeutic plans for breast cancer in the preceding two months based on existing scientific references;
- poor prognosis for survival (as judged by clinical findings).

Cardiovascular, cerebrovascular, respiratory or hematological disease and an ongoing infectious process comprised the exclusion criteria. Written consent was obtained from the patients after they had been briefed them about the benefits and effects of participating in the study. In light of the results of cell culture experiments and studies on animal models, the authorization of the university's Research Ethics Committee was also obtained. Based on evaluation by a specialist, no remedy other than narcotic drugs for pain relief were expected to benefit these patients.

HESA-A in the form of capsules was provided as a biological product of herbal and marine origin by Dr. Ahmadi to the Cancer Research Center of the Tehran University of Medical Sciences. The effective substance of the drug was passed through an 0.22 μ filter and sterilized before being encased in gelatin capsules [13]. The drug and placebo were administered respectively to those case and control patients who survived during the 6-month course of the study, at an oral dose of 50 mg/kg/day in 2–3 divided doses.

Drug intolerance at any dosage of the drug excluded the patient from the study. During the study, the patients received morphine at 20–100 mg doses depending on the severity of their pain. At the beginning and conclusion of the 2nd, 4th, 8th, 12th, 16th, 20th and 26th weeks of the study, the patients were periodically followed up for their vital signs. The patients were examined by an oncologist cooperating with the study, and the examination results were recorded.

Before initiation of drug therapy, as well as one, two, and six months after the end of the treatment period, the patients were assessed based on a pain scale, with symptoms classified as severe, moderate, little, and none.

Before drug therapy, all of the patients used narcotics 20–100 ml/day. The amount of narcotics administered to patients receiving HESA-A decreased toward the end of study. Table 1 shows the amount of narcotics used for pain relief.

Finger count and Hand Motion (HM) tests were used to assess the patients' eyesight prior to and following treatment. The average vision of patients before treatment was assessed at less than 23.400, which rose to 60.400–120.200 after treatment.

Statistical analyses were conducted using the chi-squared test, and P values below 0.05 were considered significant.

RESULTS

The patients under study had bone and retina choroid metastases and were considered end-stage. The case group underwent treatment with HESA-A, and the control group received a placebo.

Retina choroid metastasis was confirmed by an ophthalmologist prior to the study. The experience of pain was similar in both groups before treatment, as determined by the pain scale described earlier. Both groups were assessed for their

Table 1. Pain reduction scale versus duration of treatment with HESA-A (months).

Duration of treatment	0 mo.	2 mo.	3 mo.	6 mo.
Severe pain	8	5	4	1
Moderate pain	4	3	2	2
Little pain	0	3	4	5
Without pain	0	1	2	4

Pain scale: no pain – 0, little pain – 1, moderate pain – 2, severe pain – 3; ($P<0.05$).

Table 2. Comparison of the vision scale of case and control group patients before and after treatment.

Case group					
Patient number	Vision before treatment	Vision after treatment	Patient number	Vision before treatment	Vision after treatment
1	20/400	120/400	7	25/400	80/400
2	HM	40/400	8	20/400	120/400
3	40/400	80/400	9	40/400	200/400
4	14/400	120/400	10	30/400	160/400
5	5/400	120/400	11	14/400	200/400
6	30/400	160/400	12	HM	40/400
Control group					
Patient number	Vision before treatment	Vision after treatment	Patient number	Vision before treatment	Vision after treatment
1	14/400	14/400	7	HM	HM
2	HM	HM	8	10/400	14/400
3	14/400	16/400	9	14/400	14/400
4	20/400	16/400	10	2/400	2/400
5	5/400	5/400	11	40/400	40/400
6	25/400	25/400	12	30/400	30/400

Weak vision: Hand Motion (HM) = 0/400.

vision before and at the end of treatment. The patients in the control group had an average vision of 23.400 before and after the study. Vision improved notably in the case group, rising from 23.400 before treatment, to 60.400–120.400 after treatment (Table 2).

All of the patients in the case group were alive at the end of the 6-month course of the study. In the control group, however, eight, nine, and eleven patients had died by the end of the second, third, and fourth months of the study, respectively (Figure 1). Only one patient in the control group was alive by the end of the study.

DISCUSSION

This study was conducted at the Research Center of Tehran University of Medical Sciences to investigate the clinical efficacy and anticancer effects of HESA-A, a drug of herbal

marine origin, on volunteer breast cancer patients with optic metastasis and poor prognosis for survival. It was shown that 92% of the patients who received HESA-A at a dosage of 50 mg/kg/day orally lived with notably improving quality of life through the six months of the study. Reduction of the patients' skeletal pain and improved vision during the course of treatment were among the most promising results. These patients suffered fewer complications and survived longer. In general, survival is limited after the diagnosis of ocular metastasis. Ferry and Font noted a median survival of 7.4 months. Other authors have noted similar median survival rates of 5 to 20 months [16–21].

One of the numerous problems that accompany breast cancer is ocular metastasis. According to one study, the incidence of asymptomatic choroidal metastases in disseminated breast cancer was 5%, and increased to 11% when more than one organ was involved in metastatic spread [22].

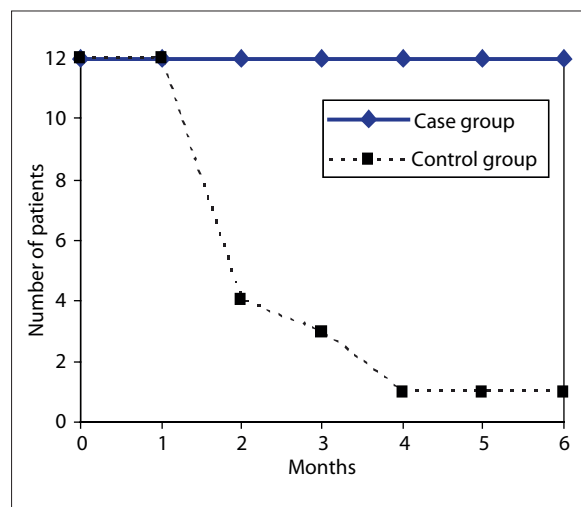


Figure 1. Survival of patients in the case and control groups.

The use of chemotherapy to treat choroidal metastasis has not been widely reported in the medical literature. Lestone et al. [23] described six patients with choroidal metastasis treated with chemotherapy. Chemotherapy can be cautiously used as management for choroidal metastasis if the primary tumor type is likely to respond to a specific chemotherapeutic regimen [24].

HESA-A is a drug which consists of antioxidant compounds, and also rare elements, such as zinc, selenium, nickel, and cobalt. The use of compounds containing zinc, niacin, vitamin C, and molybdenum has been linked to reduction of cancer mortality [25,26].

Based on cellular studies on rats and rabbits, HESA-A reaches toxic levels at concentrations of 14–16 g/kg (body weight) [13,14]. Notably, HESA-A seems to have little effect on healthy cells [15].

In this study, the intensity of the patients' experience of pain was assessed based on their use of narcotics to achieve pain relief. The use of narcotic pain killers decreased in the case group, while no changes were seen in the control group.

In the case group, ophthalmological examinations showed improvement of vision from hand motion response before treatment, to 40/400 after treatment. Vision in 7 patients improved from 5/400–40/400 before treatment, to 120/400 after treatment. No changes of vision were seen in the control group.

The results show that, despite the as yet unknown mechanism of action of HESA-A, it can bring about notable improvements, both in the patients' vision, and in their experience of pain within the experience of breast cancer.

CONCLUSIONS

In view of the great suffering of cancer patients worldwide, every likely remedy needs to be thoroughly investigated. HESA-A is a drug which has already displayed remarkable effects in reducing morbidity and increasing the lifespan of cancer patients. Further studies are warranted to understand its mechanisms of action and exploit its beneficial anticancer effects.

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